

MAT 426, Class 4: Gaussian Elimination with Partial Pivoting

Goal: Use _____ to write the matrix A as _____.

Why is this factorization important?

Definition. A nonzero square matrix P is called a permutation matrix if there is exactly _____ nonzero entry in each _____ and _____ column which is _____ and the rest are all zero.

Example 1. Consider a 3×3 matrix A and the permutation matrix $P_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$.

(a) Compute $P_1 A$

(b) Compute AP_1

Note:

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Some Important Ingredients:

- For a permutation matrix P , $P^2 =$ _____. Why? Permutation matrices are _____.

Example 2. Let $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. Compute P^2 .

- For a permutation matrix P and an elementary matrix M , $PMP =$ _____, where _____ is an _____ matrix.

Example 3. Compute the following product:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 3/4 & 1 & 0 \\ -1/4 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} =$$

- The product of two permutation matrices P_1 and P_2 is a _____ matrix.

Example 4. Compute the following product:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} =$$

Gaussian Elimination with Partial Pivoting (GEPP)

Let $A^{(0)} = A$. Then, to obtain $A^{(k)} = (a_{ij}^{(k)})$:

1.

2.

3.

4.

Example 5. Use Gaussian Elimination with partial pivoting to row-reduce $A = \begin{pmatrix} 1 & 7 & -3 \\ -3 & 2 & 0 \\ 4 & -1 & 3 \end{pmatrix}$ to an upper-triangular matrix.

Extracting P , L to get $PA = LU$

Example 6. Returning to Example 1, determine the LU decomposition of $A = \begin{pmatrix} 1 & 7 & -3 \\ -3 & 2 & 0 \\ 4 & -1 & 3 \end{pmatrix}$

(a) How do we approach this for any 3×3 matrix?

(b) What is the $PA = LU$ decomposition for the original matrix A ?